

The Specific-Factors Model

ECON 308 – International Economic Relations

Fall 2025

Two goods: Manufactures (M) and Food (F). Labor is mobile across sectors. Capital (K) is specific to M , land (T) is specific to F . Production:

$$Q_M = K^{1/2}L_M^{1/2}, \quad Q_F = T^{1/2}L_F^{1/2}, \quad L_M + L_F = L.$$

Competitive wages equal value marginal products: $w = P_M \cdot MPL_M = P_F \cdot MPL_F$.

Problem 1: Production Functions and MPLs

(a) Derive MPL_M and MPL_F . (b) Evaluate at $K = T = 100$ with $L = 100$ and $L_M = L_F = 50$.

Solution 1

$$MPL_M = \frac{\partial Q_M}{\partial L_M} = \frac{1}{2}K^{1/2}L_M^{-1/2}, \quad MPL_F = \frac{1}{2}T^{1/2}L_F^{-1/2}.$$

At $K = T = 100$ and $L_M = L_F = 50$,

$$MPL_M = \frac{1}{2} \cdot 10 \cdot 50^{-1/2} \approx 0.707, \quad MPL_F \approx 0.707.$$

Problem 2: Wage and Labor Allocation (Home)

Let prices be $P_M = 2$ and $P_F = 1$. Consider **Home** with

$$K = 144, \quad T = 64, \quad L = 100.$$

(a) State the condition for labor allocation. (b) Solve for L_M, L_F and w .

Solution 2

(a) Equalize VMPLs:

$$P_M \cdot \frac{1}{2} K^{1/2} L_M^{-1/2} = P_F \cdot \frac{1}{2} T^{1/2} L_F^{-1/2}.$$

(b) With $K^{1/2} = 12$, $T^{1/2} = 8$, $L_F = 100 - L_M$:

$$2 \cdot \frac{12}{\sqrt{L_M}} = \frac{8}{\sqrt{100 - L_M}} \Rightarrow \frac{24}{\sqrt{L_M}} = \frac{8}{\sqrt{100 - L_M}}.$$

Squaring and solving yields $L_M = 90$, $L_F = 10$. Wage:

$$w = P_M \cdot MPL_M = 2 \cdot \frac{1}{2} \cdot \frac{12}{\sqrt{90}} = \frac{12}{\sqrt{90}} \approx 1.265.$$

Problem 3: Price Change and Distribution (Home)

Keep Home endowments. Increase P_M to 3 while $P_F = 1$.

(a) Solve for the new L_M, L_F and w .

(b) Who gains and who loses among capital owners, landowners, and workers in Home?

Solution 3

(a)

$$3 \cdot \frac{12}{\sqrt{L_M}} = \frac{8}{\sqrt{100 - L_M}} \Rightarrow \frac{36}{\sqrt{L_M}} = \frac{8}{\sqrt{100 - L_M}}.$$

Square and solve: $\frac{1296}{L_M} = \frac{64}{100 - L_M} \Rightarrow 1296(100 - L_M) = 64L_M \Rightarrow L_M \approx 95.2$, $L_F \approx 4.8$.
Wage:

$$w = 3 \cdot \frac{1}{2} \cdot \frac{12}{\sqrt{95.2}} = \frac{18}{\sqrt{95.2}} \approx 1.845.$$

(b) Capital owners in M gain, landowners in F lose, workers see a nominal wage increase that is smaller than the proportional rise in P_M but larger than in P_F ; real wage rises in F units and falls in M units.

Problem 3: Two Countries, Opportunity Costs, Comparative Advantage, and Trade

Now introduce **Foreign** with endowments reversed:

$$K^* = 64, \quad T^* = 144, \quad L^* = 100.$$

Assume the same production functions.

- (c) Show that the marginal *opportunity cost* of M in units of F equals the slope of the PPF in the SF model: $OC_M = \frac{dQ_F}{dQ_M} \Big|_L = \frac{MPL_F}{MPL_M}$.
- (d) Compute OC_M for Home and Foreign at $L_M = L_F = 50$ and $L_M^* = L_F^* = 50$ using your MPL formulas.
- (e) Identify which country has comparative advantage in M and which in F .
- (f) Let the world relative price satisfy $\frac{P_M}{P_F} = \tilde{p} \in \left(\frac{2}{3}, \frac{3}{2}\right)$, e.g. $\tilde{p} = 1.2$. Who exports what under free trade? Explain briefly.

Solution 3 (Prime): Opportunity Costs, CA, and Trade

- (c) Reallocate $dL_M = -dL_F$ so $dQ_M = MPL_M dL_M$, $dQ_F = MPL_F dL_F = -MPL_F dL_M$. Then

$$\frac{dQ_F}{dQ_M} = \frac{-MPL_F dL_M}{MPL_M dL_M} = \frac{MPL_F}{MPL_M}.$$

- (d) At $L_M = L_F = 50$:

$$MPL_M = \frac{1}{2}K^{1/2}50^{-1/2}, \quad MPL_F = \frac{1}{2}T^{1/2}50^{-1/2}.$$

Hence $OC_M = \frac{MPL_F}{MPL_M} = \frac{T^{1/2}}{K^{1/2}}.$

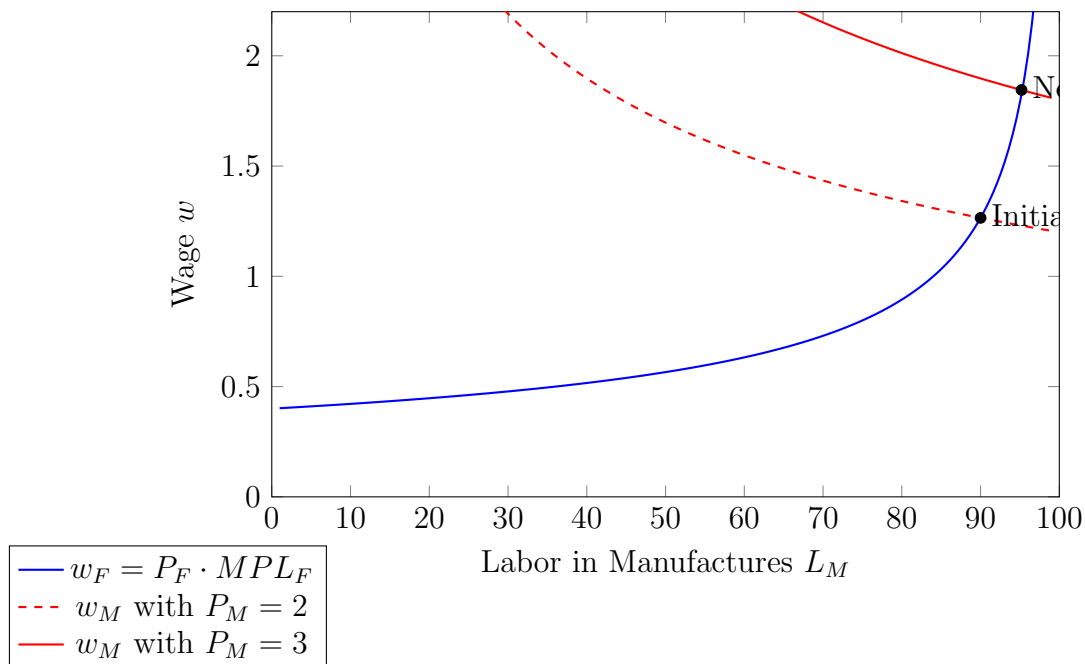
- Home: $K^{1/2} = 12$, $T^{1/2} = 8 \Rightarrow OC_M^{Home} = 8/12 = 2/3 \approx 0.667$.
- Foreign: $(K^*)^{1/2} = 8$, $(T^*)^{1/2} = 12 \Rightarrow OC_M^{Foreign} = 12/8 = 3/2 = 1.5$.

- (e) Home has the lower opportunity cost of M and thus comparative advantage in M . Foreign has comparative advantage in F . (f) With $\tilde{p} = 1.2$ satisfying $2/3 < 1.2 < 3/2$, Home exports M and imports F , while Foreign exports F and imports M .

Problem 4: Graphical Exercise (Home)

Draw $w_M(L_M) = P_M \cdot MPL_M(L_M)$ and $w_F(L_M) = P_F \cdot MPL_F(100 - L_M)$ for Home, and mark equilibria before and after P_M rises. Use parameters from Problems 2 and 3.

Solution 4: Graphical Representation for Home



Reading the figure:

- Blue: $w_F(L_M) = 4/\sqrt{100 - L_M}$.
- Red dashed: $w_M(L_M) = 12/\sqrt{L_M}$ for $P_M = 2$.
- Red solid: $w_M(L_M) = 18/\sqrt{L_M}$ for $P_M = 3$.
- Equilibrium shifts from $(L_M, w) \approx (90, 1.265)$ to $(95.2, 1.845)$.